

User Guide

for Catakig X

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updated for Catakig X by Rodney Thacker

for version 3.0
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Introduction

I really would like a chance to recreate the Apple II fun someday.

— Steve Wozniak[1]

This is the user guide for *Catakig X*, an emulator of the 8-bit family of Apple II personal computers. The Apple II was one of the most successful and influential of the early microcomputers first appearing in the late 1970s. It was also the machine that made Apple Computer famous almost overnight, and which continued to fuel the company's growth in the years before the Macintosh found its own success. Various models of Apple II were in production from 1977 all the way to 1993, making it also one of the most long-lived families of personal computers. The four models that Catakig can emulate are the Apple **][** (the original), the **][+**, the **//e**, and the **//c**.



Figure 1. The Apple **][+** and **//c** personal computers

This document assumes you already have some acquaintance with Apple II computers, though no special expertise should be needed. In particular, very little information on the Apple II's workings, its evolution, or its many capabilities is supplied in this manual, which is intended only as an operational guide to Catakig itself. However, just about all the Apple II material you could imagine existing is available these days on the Internet.

Catakig X is version 3 of Catakig. The first Catakig, released in 1997, was a classic application for PowerPC and 680x0 Macintoshes, written by Colin Klipsch.[2] Version 2, released in 2006, was a complete overhaul of the original design and a migration to MacOS X, and was published as an open-source project, intended to attract other contributors over time. Catakig X is one such contribution: version 2 brought up to date for the Macs of today by Rodney Thacker in 2026, nearly twenty years after its last release.

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[1] from a 2001 interview in the Guardian Unlimited

[2] The author will be the first to admit that the 680x0 version ran rather poorly, when it could be made to run at all.

It runs natively on Apple silicon, draws its screen with Metal, and adds a number of features that version 2 lacked. The emulation of the Apple II itself is still Colin's.

Some of version 2's absences are still absent: Catakig X has no emulated hard drives, or drives for 3½-inch diskettes. On the other hand, it can now save and restore Apple II states, and it adds emulation of the Mockingboard sound card (speech chip included), a mouse card and a clock card, game controller support, the sounds of real disk drives, and filters that make the Apple II's screen look good on a modern display. And for the **][+** and **//e**, it offers *Catakig firmware*, its own replacement for Apple's ROM, and a demonstration disk to try it on. A complete list of changes is in *What's New in Catakig X*, at the end of this guide.

Features of Catakig X

- Runs natively on Macs with Apple silicon or Intel processors, under macOS 14 (Sonoma) or later.
- Emulates the **][**, **][+**, **//e**, and **//c** models of the Apple II series.
- Runs the **][+** and **//e** either on Apple's ROM or on Catakig firmware, original replacement firmware written for Catakig X.
- Runs Apple IIs at their authentic speed of 1 MHz, or many times that speed.
- Allows multiple instances of Apple II to co-exist on your desktop, each in a separate window.
- Supports macOS's own full-screen mode, on a display of any size.
- Scales the Apple II screen sharply to any window size, with optional scanlines and NTSC composite color.
- Displays video in either monochrome or full color. Monochrome video can be in any hue of your choice.
- Displays low- and high-resolution graphics, as well as their doubled counterparts. Also renders the endearing "quirks" of high-res color graphics.
- Plays digitally filtered sound, in stereo, synchronous with the video.
- Emulates a Mockingboard C sound card in slot 2, with six voices of music and an SSI-263 speech synthesizer.
- Emulates a mouse card in slot 7, driven by your Mac's mouse or trackpad (Apple's own AppleMouse card, if you have its ROMs), and a ThunderClock-compatible clock card in slot 5 that tells ProDOS and other software the date and time.
- Saves screen images to PNG, GIF, or TIFF files. Copies the text screen, and pastes text into the Apple II as if typed.
- Emulates a pair of Disk II drives, for accessing image files of 5¼-inch disks. Supports the DSK, DO, PO, NIB, 2IMG, and WOZ formats, also inside ZIP and GZ files, and writes changes back in each disk's own format. Plays the sounds of a real Disk II while it works.
- Shows a disk image's catalog in the Finder, with Quick Look.

- Saves and restores entire Apple II states, and picks up where you left off when you next launch it.
- Reads the joystick from the mouse, the keyboard, or a game controller.
- Emulates printing: to PDF or plain ASCII text files, or to a virtual Epson dot-matrix printer, with output in PostScript.
- Comes with a demonstration disk of its own, and saves bug reports that hold everything needed to see a problem as you saw it.

Conventions of this Document

Model Names ▌ The term *Apple II* can sometimes be a little ambiguous. It is the name of the first model in the series, but it's also used as a name for the *entire* product line, without regard to specific models. This guide uses a typographic convention to keep everything clear, or to try to. The names of specific models are stylized like the emblems that appeared on their cases: e.g. the Apple **][**, or the Apple **//e**. As a name for the entire series, or for any machine when the exact model is unimportant, "Apple II" is written with a plain Roman numeral (like that one there).

Host Platform ▌ Catakig X is for Macs running macOS. Catakig 2 was written with an eye toward porting it to other operating systems, and this guide sometimes uses general language, like "host platform", out of respect for that intention. Information that's specific to macOS is identified as needed.

Editions ▌ This *User Guide* comes with Catakig X, and may be updated on a different schedule than the software, as errors are corrected and more material is added.[3] The title page will always give the guide's date of release as well as the version of the emulator it belongs to. Most of the text is Colin Klipsch's, from the guide to Catakig 2; it has been revised wherever the software has changed, and new sections describe the new features.

Acknowledgements

In the process of writing Catakig, both versions of it, I've come to owe substantial debt to the following people at the very least:[4]

Tim G.	for contributing new icons for version 2
Ron K.	for insightful discussions on Apple II programming
Nick W.	for clever tricks with Disk II emulation
Michael M.	for advice on digital audio filters
Bill N.	for many shared reminiscences on products and software
Steve W. & Steve J.	without whom writing an Apple II emulator would have been much harder, and pretty futile really.

Catakig X, in turn, owes its existence to Colin Klipsch, who wrote Catakig and released its source code for others to build on.

[3] And the weaker jokes are fixed — where possible.

[4] These are Colin Klipsch's acknowledgements, from the guide to Catakig 2.

1. Installation

In order to run Catakig for the first time, you'll initially need to do the following (though if you'll only be using the **][+** and **//e**, the second step is optional):

❶ Install the Catakig X application onto your Mac.

❷ Import one or more ROM image files corresponding to the Apple II computers and peripherals you wish to use, and that Catakig also supports. Unfortunately, Apple's ROM images **are not included** with Catakig, and must be obtained separately. For the **][+** and **//e**, Catakig X includes Catakig firmware, which stands in for them; see *Catakig Firmware*, later in this chapter.

This chapter covers these two essential steps, which are necessary when getting started. In addition, either at installation time or some later time, you might want to:

❸ Collect some disk image files containing Apple II software that interests you. These files can reside anywhere on your file system; a folder of their own in your *Documents* folder is a fine place. Catakig X comes with just one disk of its own, a demonstration disk (see *A Quick Tour*), so for anything more you'll need to bring your own software. For more about Apple II disks and disk images, and how they work with Catakig, see the *Disks & Drives* chapter.

❹ Supplement Catakig with other utilities: for example a hex editor, a PostScript viewer or PostScript-to-PDF converter, or the NuLib archiving utility. These are not absolutely critical for normal use, but can be handy to have around.[1]

Installing and Uninstalling on macOS

Catakig X runs on Macs with either Apple silicon or Intel processors, under macOS 14 (Sonoma) or later. Applying Apple's latest updates for your operating system is also recommended.

Catakig X is distributed through the Mac App Store. Install it as you would any other application from the App Store, which will put it in your *Applications* folder under the name "Catakig".

To uninstall Catakig, simply drag it from your *Applications* folder into the Trash. Catakig X runs inside macOS's *App Sandbox*, which means that everything it keeps for itself lives in a single folder in your personal *Library* folder: `Library/Containers/com.rodneythacker.CatakigX`. This holds Catakig's preferences, the ROM files you've imported, the Apple II state it saves automatically when you quit, any recovered disk copies (see the *Disks & Drives* chapter), and a log of the errors it has reported. To erase Catakig's influence completely, delete that folder too — but be aware that your imported ROM files go with it. Catakig does not install files anywhere else in your file system.

[1] Catakig can save print-outs as PDF files, but its Epson print-outs are PostScript, and since macOS 13 the Preview application no longer opens PostScript files. See the *Printing* chapter.

ROM Image Files

Like other computers, an Apple II has chips of ROM (Read-Only Memory) containing the essential machine code, or *firmware,* that lets it start up and do something useful. Among other things, Apple II ROM contains routines for printing text, drawing graphics, executing BASIC programs, computing arithmetic, beeping, and many other functions that the computer's designers knew were essential. ROM, being read-only, is immune to change from running programs. Moreover, it keeps its content even when the power is switched off. ROM therefore serves as a special storage area for data and functions considered fundamental enough to be made permanent and always available.

An Apple II depends heavily on its ROM firmware to be everything that it is. In fact, its operating systems and software won't run without it. Many peripheral and expansion cards also include firmware that's essential to their operation. From the viewpoint of you, the emulator user, it would be delightfully convenient if all this ROM content were buried away somewhere inside the emulator, out of sight and out of mind, so that you never had to think about it. There is actually no *technical* reason why this couldn't be the case. There is however a procedural reason: the ROMs are someone else's intellectual property, and cannot be freely distributed. So unfortunately, you *must* think about the subject of Apple II ROM, at the very least when you first install Catakig — unless the **][+** and **//e** are all you need, since those run on Catakig's own firmware, described at the end of this chapter. Catakig X contains no Apple ROM, and never downloads any; it uses only the Apple ROM files you give it yourself.

The content of a machine's ROM can be stored inside a file, which is then called a *ROM image file*, or just *ROM image*. Catakig keeps the ROM images you import in a "ROMs" folder of its own, and when it launches, it scans every file it finds there, looking for segments that it recognizes and can make use of.[2] In the current release, the ROM segments that Catakig knows about are:

- the 12 KB ROM of the original Apple **][**
- the 12 KB ROM of the **][+**
- the 16 KB ROM of the enhanced **//e**
- the 16 or 32 KB ROM of the various **//c** models
- the 256-byte ROM of the Disk II controller card
- the two ROMs of Apple's mouse card: the 2 KB card EPROM (341-0270-C) and the 2 KB ROM of its 6805 microcontroller (341-0269), both needed
- the files of Catakig firmware, which are described below

As far as Catakig is concerned, ROM segments do not have to be at the start of their image files, and they do not have to be the only segments in their files. It would be possible, for example, to have just one big file containing all the ROM segments you've collected. Segments do however need to be aligned within their files at positions that are

[2] Under Catakig 1, the names of the image files were significant and had to conform to a particular convention in order to be recognized. Under Catakig 2 and later this restriction has been removed.

multiples of 256, in order to be noticed. This is the usual result anyway when Apple II ROM images are created.

Importing ROM Images

Because Catakig X lives in the App Sandbox, it can't go looking around your disk for ROM files; you hand them over instead, with the *Import ROMs…* command. It's in the *Catakig X* menu, just below *Preferences…*, and there's also an *Import ROMs…* button in the *New Apple II* dialog. The very first time you launch Catakig, when it has no ROM at all, it will offer to do the importing for you.

The command brings up a dialog in which you choose one or more ROM files, or folders containing them (Catakig looks at the files directly inside a chosen folder, but not in folders within it). Each file holding at least one ROM segment that Catakig recognizes is copied into its ROMs folder; other files are ignored, and so are files Catakig has already imported. Afterward, a summary tells you which files were imported and which Apple II models now have the ROM they need.

You'll normally have no reason to visit the ROMs folder yourself, but if you want to see what's in it, or remove a file, the *Show ROM Folder* command in the *Catakig X* menu will open it in the Finder.

Acquiring ROM Images

There are a few different ways to acquire the ROM images that Catakig and other emulators need. Here are two of them:

❶ If you happen to have an Apple II ROM memorized, simply start up a hex-editor and type out all the bytes. Then save. (Rarely done.)

❷ If you own a real Apple II in good working order, boot it up and save the ROM content to a file. There is software that can do this for you, or with a bit of programming, you can do it manually. Then, transfer this file to the Mac you're going to run the emulator on, and import it.

[Details of how to create ROM images from scratch, and transfer them to your host platform, may appear in a future version of this Guide.]

Catakig Firmware

The first method above is a joke, but only partly. An Apple II's ROM is only a program, after all, and a new one can be written that does the same jobs, as long as it's written from scratch rather than copied. That is what *Catakig firmware* is: original replacement firmware for the Apple **][+** and the enhanced **//e**, written for Catakig X by Rodney Thacker without reference to Apple's code.[3] It starts the machine, boots disks, provides the Monitor and the standard routines that software calls on, and includes a BASIC that works like Applesoft, built on Microsoft's own 6502 BASIC, which Microsoft has released under an open-source license. There is Catakig firmware for the Disk II controller card,

[3] The firmware was developed under written clean-room rules: Apple's ROMs were used only as a "black box" to observe what they do, never read.

too, and for Catakig's mouse and clock cards (see *The Mouse Card* and *The Clock Card*, in the next chapter).

Catakig firmware is built into Catakig X, so there is nothing to import: a **][+** or **//e** can be created and run as soon as Catakig is installed, with no ROM files at all. It consists of five pieces, which are kept inside the application:

catakig-iiplus.rom	for the **][+**
catakig-iie.rom	for the enhanced **//e**
catakig-disk2.rom	for the Disk II controller card
catakig-mouse.rom	for the mouse card
catakig-clock.rom	for the clock card

Should a newer version of any of these files be offered separately, you can import it with *Import ROMs…*, and it takes the place of the built-in one. Catakig keeps Catakig firmware and Apple's ROM side by side, and you choose between them for each Apple II you create, with the *ROM* pop-up menu in the *New Apple II* dialog. When you have both for a model, Apple's ROM is chosen at first, and after that the dialog remembers whichever you chose last; when you have only one, that one is used. An Apple II running Catakig firmware uses the Catakig Disk II firmware as well, if it's there. There is no Catakig firmware for the original **][** or the **//c**, which still need Apple's ROM.

Most Apple II software never notices which firmware it's running on, because it uses the firmware only through its documented entry points, and Catakig firmware honors those. Some programs, though, reach into the middle of Apple's ROM for a routine that was never meant to be called from outside, or check its exact contents; copy-protected games are the usual culprits. Such programs may misbehave on Catakig firmware. If you have Apple's ROM for that model, create a new Apple II with *Apple's ROM* chosen, and run them there.

2. A Quick Tour

This chapter takes you through a short, sample session with Catakig, and provides a few helpful tips. It does not demonstrate every feature of the emulator, but should give you a good introduction to its essential operation. You should already have gone through the *Installation* chapter before proceeding any further.

First Try

Create a new Apple II. In the main menubar, choose the *New Apple II* command under the *File* menu. You should see a dialog window come up asking you to choose an Apple II model. Models for which you have no ROM will have their names tinted in red, and the *Import ROMs...* button beside the red note lets you fix that on the spot. Choose one of the allowed models, then choose the amount of extra RAM you want (though it might as well be *None* for this quick example). If you have both Apple's ROM and Catakig firmware for the model, the *ROM* pop-up menu lets you choose which one it runs. The *Mockingboard in slot 2*, *Clock card in slot 5* and *Mouse card in slot 7* checkboxes can stay as they are; there's more about them later. Catakig remembers the ROM and card choices for the next time. Then click on the *Create* button.

A new Apple II will appear, in its own window, and its screen filled with television static. That's because it has no electrical power yet.

The very first Apple II you create starts up straight away with Catakig X's **demonstration disk** in Drive 1. Its menu shows off the Apple II's graphics and sound, the Mockingboard, the mouse, clock and printer, and has a few utilities and BASIC programs to look at; choose an item by pressing its key. Everything on it was written for Catakig X, and it runs on Apple's ROM or Catakig firmware alike. To bring it back later, choose *Open Demo Disk* from the *Help* menu; Catakig asks before taking out a disk that's already in Drive 1, then restarts the Apple II with the demo disk. The demo disk lives inside the application, so it is always write-protected.

Load a disk. From the *Peripherals* menu, choose the *Load Disk Drive 1* command. A dialog sheet window will come down over the Apple II's screen, letting you select a file to load, as if the file were a real disk. Choose a disk image from your own collection — a bootable one, such as a DOS 3.3 System Master, would be ideal. (Or, simply drag the disk image's file from the Finder and drop it onto the Apple II's screen, or click the name box of Drive 1, below the screen.) After you've chosen the disk, its name should appear in the drive's name box.

Turn on the power. Press the space bar, or any regular key, to turn on the Apple II's power, or click the power button at the lower left of the window, which lights up green. The static will stop, and you should see the screen clear to black with a title at the top. The disk drive light will come on, you'll hear the drive start to whir, and the computer will begin booting the disk you loaded.

Run some programs. What happens next depends on the disk. Many disks have a start-up program which puts a menu of the disk's other programs on the screen, letting you launch them easily. Others leave you at a prompt, waiting for instructions.

At some point you may find yourself in the Apple II's BASIC interpreter, which is a command-line interface. You'll see a flashing or block cursor, with an angle-bracket (>) or square-bracket (]) character for a prompt. A full description of all the available commands is beyond the scope of this *User Guide*, but a few handy ones to keep in mind are:

CATALOG	Lists the files on a disk.
LOAD *file*	Loads a BASIC program from disk into memory.
RUN *file*	Loads, then runs a BASIC program.
BRUN *file*	Loads and runs a machine-language program (marked with a B in catalog listings).
RUN	With no file specified, runs the BASIC program currently in memory.
NEW	Erases the current BASIC program.
LIST	Lists the lines of the current BASIC program to the screen.

Apple II Windows

In Catakig, an emulated Apple II is contained in a window on your desktop. The window displays various hardware features that the real machines would typically have, all brought together as one unified package.

The Apple II's screen is the most obvious feature of the window. It occupies most of the space, and it grows and shrinks with the window, always keeping its proper shape. Below the screen, at the far left, are three buttons: the power switch, which glows green while the power is on, then *Reset* and *Reboot*, which are described below. Next to them are the disk drives. Each drive has a red indicator light (which illuminates when its motor is "on", like on the real ones), a rectangular name box that shows the name of any disk it holds, and an eject button. Clicking a drive's name box lets you choose a disk for it, and the eject button takes the disk out.

Also below the screen, and toward the right, are two more indicator lights. The one on the left, shaped like a little printer, illuminates when data is being sent to the emulated printer; click it to save what has been printed (see the *Printing* chapter). The other indicates the Apple II's current running speed, with a symbol that should look familiar: a pause sign when it's halted, a "play" arrow at normal speed, a fast-forward arrow at double speed, and a hare at six times normal speed.[1] Clicking the speed light steps to the next speed, from halted to six times normal and around again.

Finally, in the lower-right corner is a black emblem that shows the Apple II's model — which you'll probably know already, since you chose the model yourself in the *New Apple II* dialog box. But in case you forget it from time to time, the emblem is there as a reminder. Hold the pointer over it, and it also tells you whether the Apple II is running Apple's ROM or Catakig firmware.

The first Apple II window opens in the middle of your screen, a little above center. Move it or resize it wherever you like, and the next one opens in the same place. Catakig permits multiple Apple IIs to exist at the same time. Just invoke the *New* com-

[1] No tortoise is provided. The Apple II is slow enough already.

mand again to create another one. Within reason, it is possible to have as many Apple IIs as you like. However, emulation occurs only in the active or frontmost Apple II, and only when Catakig is the foreground application.

Reset, Reboot & Power

Two good menu commands to know about are *Reset* and *Reboot*, in the *Control* menu, which also have keyboard shortcuts and buttons below the screen. In general, *Reset* stops the current Apple II program immediately, and returns the machine to a known, "sane" state — though not always. (Each program can direct Reset to behave in any way it wants.) This command corresponds to the same keyboard function that the real machines have.

Reboot, also known as a "cold reset", was something the Apple **//e** and **//c** could do easily from their keyboards, but was not available on the **][** or **][+**. Catakig provides the function anyway, for all models. A Reboot acts more or less as if the Apple II had been turned off and then back on again, and forces the computer to start entirely afresh.

It's normally best to have a bootable disk already loaded into Drive 1 before invoking Reboot, since the booting of such a disk is the next thing the Apple II will try to do. If there is no disk in Drive 1, an Apple **//c** will eventually notice and alert you to the fact. The other models will simply spin the drive forever, until you either invoke Reset to stop them, or until you load a disk.

Finally, the *Power Off* command in the *Control* menu, or the power button below the screen, switches the Apple II off entirely. The screen goes black and silent, and stays that way: typing won't wake it, and Reset and Reboot are unavailable. Switch the power on again, with the same button or *Power On*, and the Apple II starts from cold, exactly as a real one does, with fresh memory and its disks still in their drives.

Typing & Pasting

Whatever you type on your Mac's keyboard goes to the frontmost Apple II, as it would on the real machine. The *Command* and *Option* keys stand in for the Apple **//e** and **//c**'s Open-Apple and Solid-Apple keys.

To type a lot of text at once, copy it on your Mac and choose *Paste Text* from the *File* menu. Catakig types it into the Apple II for you, one keystroke at a time, at whatever pace the running program is willing to read them — so a long BASIC listing can be pasted in whole. Along the way, line breaks become Returns, and on the Apple **][** and **][+**, which have no lowercase, lowercase letters become capitals. To stop a paste in progress, press any key, or Reset.

Going the other way, *Copy Screen Text* copies whatever text is on the Apple II's screen to your Mac's clipboard.

Joysticks & Paddles

Many Apple II games want a joystick or a pair of paddles. In Catakig's *Preferences* window you can choose what controls them:

Mouse ▌ The mouse's position on your screen sets the joystick's position.

Numeric keypad ▌ The keypad's number keys move the joystick: 7 8 9 / 4 5 6 / 1 2 3 push it in those directions, and it springs back to center when you let go. Keypad 0 is paddle button 0 and keypad . is paddle button 1.

Letter keys ▌ The standard choice, for Macs with no numeric keypad: while Caps Lock is off, Q W E / A S D / Z X C move the joystick the same way. Turn Caps Lock on to type those letters, for example to enter BASIC commands. Joystick keys don't type into the Apple II.

Game controller ▌ A modern game controller — Xbox, PlayStation, or one made for the Mac — connected by Bluetooth or USB. Its left stick drives paddles 0 and 1; its A (Cross) button is Open-Apple, or paddle button 0, and its B (Circle) button is Solid-Apple, or paddle button 1. Its Menu (Options) button types Escape, which pauses many games.

The Mouse Card

Later Apple II software, especially for the **//e**, could be driven by a mouse, plugged into a mouse card. A mouse card goes in slot 7, added with the *Mouse card in slot 7* checkbox in the *New Apple II* dialog. The **//c** has no slots, so it can't have one. Catakig X can provide either of two cards:

Catakig's mouse card ▌ A card of Catakig X's own design, with its own Catakig firmware built in (see *Catakig Firmware*, in the *Installation* chapter). Software that uses the mouse through the card's firmware, as programs are supposed to, works with it, including BASIC programs that read it with IN#7; software that programs Apple's mouse card hardware directly won't find it.

Apple's mouse card ▌ The AppleMouse II card itself, emulated down to its own microcontroller and running its own ROMs, so even software that bypasses the firmware works. Catakig doesn't include those ROMs; import them as you would Apple's other ROMs (see *ROM Image Files*), and the import summary tells you when both are present.

An Apple II running Apple's ROM gets Apple's mouse card when its ROMs are installed, and Catakig's otherwise; one running Catakig firmware always gets Catakig's.

When a program turns the mouse on, your Mac's pointer disappears over the Apple II's screen, since the program draws its own; moving your mouse or trackpad then moves the Apple II's pointer, and clicking is its button. To get your Mac's pointer back, push the Apple II's pointer past an edge of its screen (in full screen, the top edge, toward the menu bar), or switch to another application. If the program keeps its pointer from reaching the edge, just keep pushing in the same direction, and your Mac's pointer comes free. Your pointer is always free while a menu is open, and stays free afterward until you move it back over the Apple II's screen and click. With a mouse card installed, the mouse drives the card rather than the joystick.

The Clock Card

An Apple II has no idea what time it is unless a clock card tells it. Catakig X emulates one in slot 5 that works like Thunderware's ThunderClock Plus, the card that ProDOS and most Apple II software know how to read, with Catakig firmware of its own in place of Thunderware's. It keeps your Mac's local date and time, so ProDOS stamps the files it saves, and programs that show the time show the right one. Like the other cards, it's added when an Apple II is created, with the *Clock card in slot 5* checkbox in the *New Apple II* dialog, and the **//c**, which has no slots, can't have one.

From BASIC, the clock is read the ThunderClock way: send output to it with `PR#5` and input from it with `IN#5`, then `INPUT` a string. The first character sent to the card picks the format: `%` gives a 12-hour time such as "TUE MAY 12 4:32:55 PM", `&` the same with a 24-hour time, and `#` just the numbers. A program that sets the clock changes the Apple II's time only, never your Mac's: the card simply runs that much ahead of or behind the Mac from then on.

Saving & Resuming

Catakig X can save the complete state of an Apple II — its memory, its CPU, its screen, its Mockingboard, and which disks are in its drives — to a file, and open it again later exactly as it was. Use *Save Machine State* and *Save Machine State As…* in the *File* menu to save, and *Open Machine State…* to open one. Saved states are files with names ending in "a2st", and double-clicking one in the Finder opens it too.

You don't have to remember to save, though. When you quit Catakig, it saves the frontmost Apple II automatically, and restores it the next time you launch Catakig, so that you can pick up exactly where you left off.[2] When you close an Apple II's window yourself, Catakig asks whether to save its state first.

A saved state remembers its disks by reference, not by copying them. If you move or rename a disk image after saving, Catakig won't be able to find it when the state is opened, and will ask you to load it again. And since a state depends on the ROM the Apple II was running, Catakig warns you if you open a state saved with a different ROM than the one installed now. A state remembers whether it ran Apple's ROM or Catakig firmware, too, and runs on the other only if that one is no longer installed.

Some Performance Tips

Catakig emulates computers from around 1980 that had 8-bit, 1-MHz microprocessors. You would think, surely, that your modern machine with its blazing clock rate thousands of times faster should have no trouble emulating a humble Apple II. And you'd be right. Desktop computers have long been fast enough to emulate an Apple II at full speed.

Nevertheless, Catakig still asks some work of your computer. Not only must it emulate approximately 300,000 instructions of the 6502 CPU every second, but in addition it must render the video frames that you see and the sound waveforms that you hear.

[2] Even in the middle of a game. Especially in the middle of a game.

Moreover, Catakig allows Apple IIs to run at their authentic 1-MHz speed or at many times that speed, which multiplies the demands on the host platform.

Great pains[3] have been taken to make Catakig as efficient as possible, and any Mac that can run macOS 14 has power to spare for it. However, if you do find Catakig's performance underwhelming, or you'd like your battery to last longer, consider one or more of the following remedies:

- As mentioned, Catakig can emulate Apple IIs at many times their authentic speed. The price you pay for exercising this feature is a higher demand on the host system's CPU. (The "halted" speed is the most economical of all, though not nearly as much fun.)
- The *Composite Color* display filter does more work than the others. If animation looks less than smooth, try turning it off.
- When Catakig is the foreground application, other open applications will be in the background — possibly doing something time-consuming, possibly not. Those that often do include: backup and file utilities, video encoders, live-action games, and web browsers with many busy tabs. When you run Catakig, try quitting such applications, or at least pausing their activities, to stop them from hogging your computer's resources. Let Catakig do that instead.

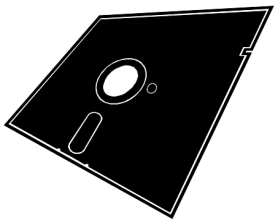
Reporting a Problem

If a program misbehaves, or Catakig itself does, choose *Save Bug Report…* from the *Help* menu while the problem is still on the screen. The Apple II pauses, and Catakig asks you to describe in a sentence or two what happened and what you expected, and where to save a ZIP file. Send that file to the developer with your report; Catakig never sends anything by itself.

The report holds the Apple II's complete state, which can be opened to see exactly what you saw, along with a picture of its screen, a summary of the machine, the last instructions it ran, the errors and warnings Catakig has shown lately, and the versions of Catakig and macOS. Code from Apple's ROM is left out of its text. Of your own files, it names only the disk images in the drives, and includes the disks themselves only if *Include the disks in the drives* is checked. Leave it checked unless the disks are private, since without them most problems can't be reproduced.

[3] You have no idea.

3. Disks & Drives



Every Apple II in Catakig comes with a pair of emulated *Disk II* disk drives attached and ready for use. They are represented at the bottom of the Apple II's window: each one has a rectangular display to show the name of the disk it holds, and an indicator light that illuminates when there is disk activity.

Along with the pretend disk drives go pretend disks, which for Catakig and many other emulators must take the form of files on your host platform. This chapter discusses what these files are and how they are used.

Disk Image Files

As is the case for ROMs, the contents of Apple II disks can be captured and stored in files. These files then serve as equivalent substitutes for the real disks, at least as far as the emulator is concerned. The Apple II had several popular forms of disk media, among them: ❶ the ubiquitous 5¼-inch "floppy" disks, with a capacity of 140 KB; ❷ 3½-inch disks with a capacity of 800 KB; and ❸ external hard disks with capacities of many megabytes. The current release of Catakig supports only the first of these.

If you have an Apple II with the right hardware, there is software you can run to make image files out of real disks. The details of this process are not discussed in this guide however. You can also take advantage of the fact that other people have been making disk images for a long time and then putting them on the Internet.

Unfortunately, the image files for representing Apple II disks come in a bewildering variety of standardized formats.[1] The ones that Catakig recognizes are:

DO A 5¼-inch disk, stored with DOS sector ordering. Might or might not be formatted by DOS.

PO A 5¼-inch disk stored with ProDOS block and sector ordering. Might or might not be formatted by ProDOS.

DSK Exactly like either DO or PO, but you'll have to guess which one. In practice, it's usually DO.

NIB A 5¼-inch disk, stored in the expanded *nybblized* encoding used by the Disk II controller card. These files are about 230 KB in size, even though they still represent only 140 KB worth of content. The filenames are usually suffixed with "nib", or sometimes "nyb". A rarer, slightly smaller variety, with 6,384 bytes per track instead of 6,656, is also accepted.

2IMG An inclusive file format that covers DO, PO, and NIB, as well as larger disk volumes. Filenames are usually suffixed with "2img" or "2mg". Catakig loads 2IMG files that hold 5¼-inch disks.

WOZ A newer format that records a disk's magnetic patterns, bit by bit, as captured from the original by special hardware. Catakig reads both versions,

[1] "The wonderful thing about standards is that there are so many to choose from." — Grace Hopper, or possibly Andrew Tanenbaum

WOZ 1 and WOZ 2, though not the timing details that some copy protection depends on. WOZ disks always load write-protected.

Disk images are often passed around compressed, and Catakig can load any of these formats straight from a **ZIP** or **GZ** file, without your having to expand it first. When a ZIP file holds more than one disk image, as collections of multi-disk games often do, Catakig asks which one to load. Since changes can't be written back into a compressed file, disks loaded this way are write-protected. (ShrinkIt archives and other Apple II archive formats still need expanding first.)

Catakig X is more careful than its predecessors about telling these formats apart, especially 2IMG files. It also recognizes images of 3½-inch disks, including Apple's Disk Copy 4.2 format, but since it has no 3½-inch drive to put them in, it can't load them yet.

Loading & Unloading Disks

To **load** a disk image — as if you were physically inserting a real disk into a real drive — choose one of the *Load Disk Drive* commands from the *Peripherals* menu. A dialog box will come down over the Apple II's window and allow you to choose a file. Only valid disk image files should be selectable in this dialog. If the disk loads successfully, its name will then appear in the drive's rectangular display.

There are quicker ways too. Click a drive's name box to choose a disk for that drive. Drag a disk image from the Finder and drop it onto a drive's name box to load it into that drive, or onto the Apple II's screen to load it into Drive 1 (hold down the *Option* key for Drive 2). Double-clicking a disk image in the Finder, or dropping it on Catakig's icon in the Dock, loads it into Drive 1 of the frontmost Apple II.

In general, Drive 1, on the left, is the main drive, and is the one that disks can be booted from. Drive 2, on the right, cannot be used for booting but is otherwise the same.

As for **unloading**, you'll normally have no need to actively remove disks from their drives. Disks are automatically unloaded whenever newly loaded disks replace them, or when an Apple II is closed. However, if for whatever reason you do want to empty a drive, click the eject button beside its name box, or use the *Unload Drive* commands in the *Peripherals* menu.

Disk II drives were never quiet, and Catakig's aren't either: you'll hear the motor whir, the head step from track to track, the familiar rattle as it recalibrates against track 0, and the door as disks go in and out. These are recordings of a real Disk II. If they get to be too much, turn them off with the *Disk drives* checkbox in the *Preferences* window.

The moment of unloading is a significant one for write-enabled disks that have been modified during their time inside a drive. When a modified disk is unloaded, the changes are written back into its image file, *in the file's own format*: a DO file stays a DO file, a 2IMG file keeps its header, and the file's name is left alone. Only the tracks that were changed are written. (If the disk hasn't been modified, its image file will remain untouched.) This writing occurs only at unloading time, and not while the disk is actually being written to under emulation.[2]

There is one catch. A DO, PO, or 2IMG file stores only the *data* of a disk's sectors, so the changes can be written back only if every changed track still holds standard, readable sectors. Copy-protected software, and some programs that format disks in their own way, can write tracks that don't. When that happens, Catakig leaves the original file untouched, and asks you what to do instead:

Save Copy as .nib... ▌ Saves the whole disk, changes included, as a new NIB file, which can hold anything a real disk can.

Discard Changes ▌ Throws the changes away.

Cancel ▌ Leaves the disk in the drive, as it is.

If the question can't be asked — when you quit Catakig with several Apple IIs open, for instance — Catakig saves a NIB copy for you, in a "Recovered Disks" folder inside its own folder (see *Installing and Uninstalling*). The disks in the frontmost Apple II are not unloaded when you quit at all: they stay in its drives, in the state that Catakig saves automatically, and are written back when they are eventually unloaded.

Write-protected Disks

Real Apple II disks have ways of indicating that their contents are to be protected from modification. The 5¼-inch disks have a write-enable notch on one side that can be covered with a piece of black tape.[3] The 3½-inch disks have a small square slider that can be pushed to one side, or removed entirely, to indicate write-protection. In either case, the drive will sense the physical feature that means "write-protected" and then prevent the computer from changing any data on the disk.

Disk image *files* don't have these physical features of course. Instead, Catakig goes by the operating system's permissions for the file: if the file cannot be opened for writing, then Apple IIs will see it as a write-protected disk.[4] The color of the disk's name, as displayed in the drive, will indicate its status. **Green** means the disk is writeable; **yellow** means it's write-protected.

Note though that Catakig's assessment of a disk's write-protection is done *only* at the moment it is being loaded. It does no good, for example, to change the image file's attributes while the disk is in use, after you notice it's in the wrong state for your purposes. In general, and when in doubt, it's a good idea to keep your disk images write-protected by default, and only write-enable the ones that need it.

Catakig supports multiple Apple IIs existing at once, and it is possible to have the same disk image loaded into more than one Apple II at the same time, or even loaded into both drives of the same Apple II at the same time — a strange situation that cannot arise on the real machines. Under these circumstances, a write-enabled disk will be

[2] Catakig 2 did things differently: it rewrote a modified disk as a NIB file and changed its filename suffix to "nyb". Catakig X never changes a disk image's format or name.

[3] Or in a pinch, a small wad of chewing gum. Preferably licorice.

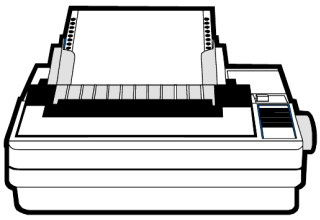
[4] On macOS, the file can also be *locked*, which is a separate attribute from the traditional Unix file permissions, but which also has the effect of write-protection. A 2IMG file can also mark itself as locked in its header, and Catakig honors that too.

writeable in *at most* one of the drives it's loaded into, the one that last opened the file for writing and succeeded. In all others, the disk will appear write-protected.

Looking Inside Disk Images

Catakig X adds a Quick Look extension to the Finder. Select a disk image in a Finder window and press the space bar, and Quick Look shows the disk's catalog: the DOS 3.3 CATALOG, a ProDOS directory (subdirectories included), or an Apple Pascal directory, with each file's type and size, and the free space. The disk's icon in the Finder shows its catalog too, on a little green screen. Even a copy-protected disk shows whatever can be read from it.

4. Printing



Catakig does not support printing to any *real* printers attached to your host computer. This is just as well, since modern printers do not have control languages that old Apple II software would normally know how to speak.

Instead, Catakig provides emulated printing (naturally) which renders output to files. Other applications can then print these files onto actual paper as desired. File print-outs have the advantage that they can be printed multiple times, and stored indefinitely, and shared with others over the Internet.[1]

Print Sessions in Catakig

When you print things on a *real* Apple II, the computer sends a stream of bytes to its attached printer, which interprets the bytes immediately as they arrive and responds to them appropriately. The printer marks the paper with characters, advances the printhead forward, scrolls the paper, and so on.

Under Catakig, the chain of cause and effect is more indirect. Apple II programs running under emulation will "believe" they are sending data to a printer, but the bytes are actually accumulating in a hidden buffer called a *print session*. The print session just sits there, passively, until you decide to do something with it. When your emulated printing has finished, or when it reaches a good stopping point, you might then want to **save** the session to a print-out file.

When you do save it, you'll have a choice of how to process the collected bytes to turn them into an output file, based on the sort of printer you're pretending to have, and the format of the file. The current release of Catakig provides the following formats:

PDF (text only) ▌ The standard choice. The text is laid out as an 80-column printer would put it on US Letter paper, ten characters to the inch and six lines to the inch, sixty lines to a page, with form feeds starting new pages. The output is a PDF file, which any Mac can open and print. Printer control codes and graphics are left out.

Plain ASCII ▌ The data stream is assumed to be for a plain ASCII printer with no graphics or other fancy features — such as the old *daisy-wheel* printers. The output will be an ASCII text file.

Epson to PostScript ▌ The data is assumed to be intended for an Epson RX-80 dot matrix printer, and is interpreted accordingly. The data may contain Epson control sequences, or it may be just ordinary ASCII. The output will be a PostScript file that resembles the output from real Epson printers.

Raw bytes ▌ The data stream is not interpreted at all, but simply copied as-is to the output file you specify. This is normally not a very useful option, but might be of interest if you want to capture the raw byte stream and examine it.

[1] Another big advantage: they let the author put off support for real printing.

Saving a session begins when you invoke the *Save Print Session...* command from the *Peripherals* menu, or simply click the printer light below the screen. A dialog window will appear over the Apple II window and allow you to name the destination file, as well as choose its format. The dialog also shows the number of bytes currently sitting in the session, to give you a rough idea of how large the output file will be. If nothing has been printed yet, Catakig says so instead. Most Apple II software prints to slot 1, which is where Catakig's printer is; in BASIC, `PR#1` sends output to it, and `PR#0` sends it back to the screen.

Print sessions are not affected by being saved to an output file. In particular, they are not automatically erased. The data in the session will still be there, and any additional data you send later will continue accumulating on top of the old. In order to reset a print session back to its original, empty state, you need to **clear** it, via the *Clear Print Session* command in the *Peripherals* menu.

Each Apple II in Catakig keeps its own private print session, which is created and initialized along with the computer. Any printing activity taking place in one Apple II has no effect on the printing in others.

Viewing PostScript Print-outs

Catakig's PostScript files used to open directly in Apple's *Preview* application. Since macOS 13, unfortunately, Preview no longer opens PostScript at all. For text, the simplest answer is the *PDF (text only)* format, which opens in Preview directly. For an Epson print-out with graphics or special type styles, save it as PostScript and convert it to PDF; the free Ghostscript package, for one, includes a converter called `ps2pdf` that does exactly that.

5. Video & Sound

An Apple II was usually connected to whatever screen was at hand: a small monochrome monitor, if its owner was lucky, or the family television, if not. Its sound came from a little speaker inside the case, which could do nothing but click. This chapter describes how Catakig X recreates both, on hardware that the Apple II's designers never imagined, and how you can add a sound card to the mix.

Display Filters

The Apple II's screen is just 280 dots wide and 192 lines tall, and your Mac's display has many times that number of pixels. How the one is stretched over the other makes a surprising difference, so Catakig X offers a choice, under *View › Display* in the menubar. The choice applies to all Apple II windows at once.

Sharp Pixels ▌ Every Apple II dot is drawn as a crisp, solid block, the same size as all the others, with just enough smoothing at the edges to keep them even at any window size. This is the standard setting.

Scanlines ▌ Draws thin dark gaps between the lines of the screen, as a real monitor did. Scanlines need room to show, so they appear only when the screen is at least three times its original height; in a smaller window, you'll get Sharp Pixels instead.

Composite Color (NTSC) ▌ This one is a switch rather than a choice, and works together with either of the others. It decodes the Apple II's video signal the way a color television or composite monitor would, so that colors blend and bleed into each other as they did on the real thing, and dithered high-res patterns turn into the solid colors their artists intended.

Separately, the *Monochrome/Color* command in the *Window* menu switches an Apple II between a color display and a monochrome monitor. The monochrome hue (green, as it happens, until you decide otherwise) is set in the *Preferences* window.

Full Screen

The *Enter Full Screen* command in the *View* menu (Shift-Command-F), or the green button at the top of the window, gives an Apple II a whole display to itself, using macOS's standard full-screen mode. Catakig never changes your display's resolution; instead, the Apple II's screen is scaled up to fill it, keeping its proper shape, and the mouse pointer is hidden while it's idle. Choose *Exit Full Screen* to return to the desktop.

In full screen, the bar below the Apple II's screen, with its power button, drives and lights, is tucked out of sight. Choose *Show Toolbar* from the *View* menu, or press Shift-Control-Z, to slide it up. It slides away again when you press the keys a second time, click anywhere else, or leave it alone for five seconds. While it's up, your Mac's pointer is free to use it, even if a program has the Apple II's mouse turned on.

Screen Images

The *Save Screen Image* command in the *File* menu saves a picture of the Apple II's screen, as you see it, to a PNG, GIF, or TIFF file.

Sound

Catakig plays the Apple II's sound through your Mac's current sound output, and you set its loudness the way you set any other application's, with your Mac's volume controls. Catakig X produces its sound in stereo, and the sound drives the timing of the emulation, which keeps both the music and the Apple II itself running at a steady pace. When an Apple II is halted, or its window isn't frontmost, it falls silent.

Along with the Apple II's own sound, Catakig X plays the sounds of its disk drives, as described in the *Disks & Drives* chapter, and the clunk of the power switch. The *Sounds* checkboxes in the *Preferences* window, *Disk drives* and *Power switch*, turn each of these off or on.

The Mockingboard

The Apple II's speaker was famously limited, and in 1983 Sweet Micro Systems did something about it. Their *Mockingboard* was a sound card with two General Instrument AY-3-8910 sound chips, giving six voices of music and noise in stereo, and the Mockingboard C model added a Votrax SSI-263 chip that could speak. Many games and music programs were written to use it.

Catakig X emulates a Mockingboard C, installed in slot 2. Whether an Apple II gets one is decided by the *Mockingboard in slot 2* checkbox in the *New Apple II* dialog, which is normally checked; Catakig remembers your choice for next time. The Apple **//c** has no slots, so it can't have a Mockingboard, and the checkbox is disabled for it. The Mockingboard is saved and restored along with the rest of an Apple II's state.

Most Mockingboard software searches the slots to find the card, wherever it is. A few programs insist on finding it in slot 4, which in Catakig is occupied by a memory card, and those won't find it.

As for speech, the SSI-263 built its words out of *phonemes*, the individual sounds of spoken language, and Catakig X synthesizes those sounds itself. The result is recognizably speech, and recognizably robotic, as it was on the real card; but it's an approximation, and doesn't sound exactly like the original chip.[1]

[1] The original chip didn't sound exactly like a person either.

What's New in Catakig X

Catakig X, version 3.0, is the first release since Catakig 2.00b4 in October 2006. These are the changes, roughly in order of how much you're likely to notice them.

- **Runs on today's Macs.** Built as a universal application for Apple silicon and Intel processors, for macOS 14 or later, with the old PowerPC and Mac OS X 10.3 support removed.
- **New video.** The screen is drawn with Metal, scales to any window size, and offers the Sharp Pixels, Scanlines, and Composite Color (NTSC) display filters.
- **Standard full screen.** Full screen now uses macOS's own full-screen mode, without changing the display's resolution.
- **Reworked sound.** Sound is now produced in stereo and drives the emulation's timing, for steadier sound and a steadier speed.
- **Catakig firmware.** Original replacement firmware for the **][+**, the enhanced **//e**, and the Disk II card, built in, so the **][+** and **//e** run without any Apple ROM; Apple's ROM, if you import it, can be chosen per Apple II instead.
- **Mockingboard C** sound card in slot 2, with SSI-263 speech.
- **Mouse card** in slot 7, driven by the Mac's mouse or trackpad: Catakig's own, or Apple's AppleMouse II card running its own ROMs, if you import them.
- **Clock card** in slot 5, compatible with the ThunderClock Plus, keeping the Mac's date and time for ProDOS and other software.
- **A demonstration disk,** built in, which the first Apple II starts with and *Help › Open Demo Disk* brings back.
- **More disk formats.** WOZ disks, and disk images inside ZIP and GZ files, with a choice of disk when a ZIP file holds several.
- **Disk drive sounds,** recorded from a real Disk II, and a power switch sound.
- **A power switch,** and Reset and Reboot buttons, below the screen; eject buttons for the drives, whose name boxes can be clicked to load a disk; and a speed light that can be clicked to change speed.
- **Quick Look** previews and icons showing a disk image's catalog in the Finder.
- **Saved states.** Apple II states can be saved and opened, and the frontmost Apple II is restored automatically at launch.
- **Disk write-back in the original format.** Modified disks are written back into their own image files, in their own format, instead of being converted to NIB files and renamed. Changes that can't be stored that way are offered as a NIB copy, never forced onto the original.
- **Better disk format recognition,** for 2IMG files in particular, and for Disk Copy 4.2 images (which can't be loaded yet).
- **Printing to PDF.** Print sessions can be saved as PDF files, now the standard format, and clicking the printer light saves them.

- **Bug reports.** *Help › Save Bug Report…* saves the Apple II's state, its screen, its recent instructions and Catakig's recent errors in one file, to send to the developer.
- **Paste Text fixed.** Pasting now works reliably, for text of any length, with line endings and letter case converted for the Apple II.
- **New indicator lights** for the drives, the printer, and the speed, drawn sharply at any resolution.
- **Game controllers** can be used as the Apple II's joystick and buttons, with the Menu button as Escape.
- **Drag and drop** of disk images onto Apple II windows, and opening disk images from the Finder.
- **Importing ROMs.** Catakig X runs in macOS's App Sandbox and includes no Apple ROM; the *Import ROMs…* command brings in your own ROM files instead of a ROMs folder next to the application.
- **Help menu.** This guide, a Troubleshooting & FAQ page, a list of keyboard shortcuts, and the license and legal notices are all in the *Help* menu.
- **Remembered choices.** The *New Apple II* dialog remembers its ROM and card choices, and the Apple II window reopens where you left it.

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Catakig

Catakig
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Catakig X

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Catakig Firmware

Catakig firmware
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